

Upper Respiratory Tract Infections With Focus on The Common Cold

Authors

Ellis H. Tobin¹; Micah Thomas²; Paul A. Bomar³.

Affiliations

¹ Albany Medical college

² Grand Strand Regional Medical Center

³ Grand Strand Medical Center

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Continuing Education Activity

Upper respiratory tract infections (URTIs) are the most common infectious disease cause for patient evaluations worldwide. URTIs are typically self-limited and mild to moderate in severity. The most common manifestation of URTIs is the common cold. Due to overlapping symptoms of various viral and bacterial infections that can result in URTIs, making the diagnosis of the common cold is a challenge. Recent developments in rapid diagnostic tests have significantly impacted the ability to establish the etiology of URTIs; however, essential aspects of test sensitivity, specificity, and costs should guide their use. URTIs are often treated with unnecessary antibiotics. Therefore, clinicians and patients must be educated about the potential risks associated with antibiotic therapy.

This course explores the impact of URTIs, which, despite being a mild illness, significantly affect public health, leading to lost productivity and unnecessary medical visits. Proper patient education and adherence to evidence-based management strategies are essential in reducing the burden of URTIs. This activity for healthcare professionals is designed to enhance the learner's competence in identifying URTIs and their etiologies with a focus on rhinovirus and the common cold, performing the recommended evaluation, and implementing an appropriate interprofessional approach when managing this condition.

Objectives:

- Assess the symptoms and signs of self-limited upper respiratory tract infections.
- Apply best practices when obtaining diagnostic tests for upper respiratory tract infections.
- Select the appropriate course of therapy for upper respiratory infections.
- Apply effective strategies to improve care coordination among interprofessional team members to reduce the prescription of unnecessary antibiotics

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Introduction

Upper respiratory tract infections (URTIs) encompass a broad topic that involves a large variety of pathogens. The ubiquitous clinical syndrome recognized as the "common cold" and the rhinovirus, its most frequent etiology, will be focused on here. Other names for the common cold are acute nasopharyngitis and acute coryza. In addition to URTIs, rhinovirus can cause infection of the lower respiratory tract, has been implicated in the exacerbation of asthma and chronic obstructive pulmonary disease (COPD), and may act as a fellow pathogen in the development of viral and bacterial infections involving both the upper and lower respiratory tracts.[1][2]

The common cold is a syndrome that manifests as a mild-to-moderate, self-limited URTI, characterized by inflammation that causes malaise, nasal congestion, rhinorrhea, sneezing, pharyngitis, cough, and occasional fever and headache. Various other respiratory viruses and bacteria, mentioned here but not presented in detail, can cause

common cold symptoms that overlap with those due to rhinovirus. Please see StatPearls' companion resources, "[Pharyngitis](#)," "[Acute Laryngitis](#)," "[Acute Sinusitis](#)," "[Recurrent Acute Rhinosinusitis](#)," "[Acute Otitis Media](#)," "[Mastoiditis](#)," "[Influenza](#)," "[Parainfluenza Virus](#)," "[Respiratory Syncytial Virus Infection in Children](#)," "[Adenoviruses](#)," "[Human Metapneumovirus](#)," and "[Features Evaluation and Treatment of Coronavirus COVID19](#)," for further information on these pathogenic etiologies.

URTIs are the most common acute infectious diseases in the world.[1] In the vast majority of cases, they tend to be self-limited but, on occasion, can evolve into chronic and significant complications. They invariably impact the quality of life, cause absence from school and work, and substantially contribute to the prescription of unnecessary antibiotics. In the United States, the annual economic burden of URTIs is estimated to be 60 billion dollars.[3]

Etiology

The following respiratory viral pathogens (see **Image**. Classification of URTIs With Associated Most Relevant Causative Agents) can cause symptoms of the common cold:

- Rhinovirus (responsible for 50% to 80% of cases)
- Coronavirus
- Adenovirus
- Parainfluenza virus
- Influenza virus
- Respiratory syncytial virus
- Enterovirus
- Human metapneumovirus
- Bocavirus

Other viruses that can cause symptoms of the common cold include:

- Measles
- Enterovirus

Additionally, the following bacteria represent a partial list of pathogens that can cause and contribute to URTI symptoms:

- Group A beta-hemolytic *Streptococcus* (*Streptococcus pyogenes*)
 - Group C and G beta-hemolytic Streptococci
- *Mycoplasma pneumoniae*
- *Moraxella catarrhalis*
- *Haemophilus influenzae*
- *Streptococcus pneumoniae*

Rhinovirus is a non-enveloped positive-sense single-stranded RNA virus belonging to the *Picornaviridae* family.[2] [[Noninfluenza Respiratory Viruses](#)] It consists of over 160 antigen subtypes. Multiple subtypes can circulate in communities simultaneously, and individual subtypes have not been definitively linked to differences in illness severity or seasonality.

Epidemiology

In 2021, the global number of all-cause URTIs (excluding COVID-19) was 12.8 billion for all age groups, with an incidence rate of 162,484 per 100,000 population. Children younger than 2 years had the highest rates, and children between the ages of 5 and 9 had the highest number of episodes. The mortality rate (excluding COVID-19) was 0.2 per 100,000.[4] While the specific epidemiology statistics related to rhinovirus are unknown, it is reasonable to assume that a large proportion of the URTI burden is due to the virus.

Rhinovirus circulates worldwide, and infections occur throughout the year, with the peak of clinical infections occurring in the fall and spring.[5] In contrast, the incidence of respiratory infections due to influenza, respiratory syncytial virus (RSV), and coronavirus peaks in the winter. Seasonality may be influenced by humidity and temperature, although definitive meteorological mechanisms have not been fully established.[6] The severity of rhinovirus infections increases in the winter, probably due to alterations in host defense mechanisms.

[6] Asymptomatic infection throughout the year is common.[7]

Rhinovirus is transmitted via several routes, including aerosols and large droplets, through contact with contaminated surfaces, followed by self-inoculation of nasal and conjunctival mucosa. Whether the aerosol route or direct contact is most responsible for transmission is uncertain. The virus can linger on surfaces for several days, easily transferring the pathogen to fingers.[8] The incubation period is approximately 1 to 4 days. Psychosocial factors, including stress, depression, sleeplessness, smoking, and poor nutrition, appear to increase susceptibility to rhinovirus-associated URTIs, likely due to impaired immunologic responses.[1]

The incidence of the common cold in the general adult population is between 2 to 5 episodes per year. Children represent the reservoir of rhinovirus, and the incidence rate of infection is approximately 4-fold higher than that of adults, likely due to child-to-child contact at daycare and school settings.[9] Consistent with the transmission mechanisms of respiratory viruses, a direct correlation between crowding and the risk of developing a URTI is present. A correlation between URTIs and the quality of indoor ventilation is not apparent.[10] Exposure to cold air, originally thought to be the harbinger of the syndrome that carries its name, does not predispose individuals to catching the common cold.

Pathophysiology

The rhinovirus incubation period ranges between 1 to 4 days. After inoculation of the nasal or conjunctival mucosa, the virus can be recovered in the nasopharynx. In contrast to influenza-induced epithelial cell damage, rhinovirus infection does not cause direct cell cytotoxicity. In the latter case, symptoms of the common cold are a result of innate immune response.[1] Rhinovirus infection induces an increase in inflammatory mediators and cytokines. Bradykinin production is associated with the onset of a sore throat, rhinorrhea, and nasal congestion.[11]

Nasal congestion results from the dilation of the veins within the sinusoids of the nasal turbinates. It has been postulated that the increase in nasal blood flow may be an antiviral defense mechanism whereby the resulting increase in intranasal temperature prevents further viral replication.[12]

The mechanism underlying the cough associated with the common cold is poorly understood. Experts have theorized that the sensitivity of airway sensory nerves increases due to increased bradykinin and tachykinin production. [13] Cough may also result from rhinovirus infection of the lower respiratory tract, as well as irritation of the pharynx from nasal mucous secretions.

Cytokines and prostaglandins likely contribute to the systemic symptoms of malaise, muscle aches, and headache. [1] Rhinovirus can spread to the lower respiratory tract, bloodstream, and gastrointestinal tract. Furthermore, rhinovirus can be a co-pathogen with other respiratory viruses and bacteria.[2]

Infections by rhinovirus produce neutralizing humoral responses with the production of serotype-specific IgG and IgA antibodies. The antibody responses are not fully protective against reinfection with the same serotype. In addition, minimal cross-reactivity of the neutralizing antibodies occurs between the large number of rhinovirus serotypes. [14] Besides rhinovirus, the other respiratory viruses also contribute to the recurrent nature of the common cold.

History and Physical

Clinical History

In the vast majority of cases, common cold symptoms are recognized by the patient within whom they have manifested and are not brought to the attention of the healthcare team. That said, given the sheer magnitude of URTIs in the general population, those individuals who do seek medical attention represent a large segment of the population. Patients typically present to clinicians seeking symptom relief.

The history should account for recent ill contacts, exposure to daycare, school, residential, and work settings, recent travel, underlying comorbidities, vaccinations, and allergies. The review of symptoms should attempt to establish a semi-quantitative severity score of pharyngeal, sinus, ear, and headache pain. Characteristics of the cough should be described, and the presence or absence of dyspnea should be identified.

Individual symptoms of the common cold can vary in severity and duration. In general, and accepting their subjective nature, symptoms tend to be mild to moderate in severity. Sore throat is often the earliest symptom, followed 1 to 2 days later by rhinorrhea and sneezing. Nasal congestion, laryngitis, and cough typically develop shortly thereafter. Malaise, low-grade fever, myalgias, and headache are common during the early phase of infection. Symptoms typically last 7 to 10 days, although the cough can linger for several weeks.

Physical Examination

Vital signs are typically within normal limits. If present, fever is usually low-grade. Clinical findings of a high fever, pharyngitis, tonsillitis, otitis, cervical adenopathy, wheezing, productive cough, or dyspnea add complexity to the diagnosis and warrant further investigation to determine whether complications from the common cold have developed or an alternative diagnosis exists. Please see StatPearls' companion resources, "[Pharyngitis](#)," "[Acute Otitis Media](#)," "[Acute Sinusitis](#)," "[Acute Laryngitis](#)," "[Asthma](#)," "[Bronchiolitis](#)," "[Acute Bronchitis](#)," and "[Viral Pneumonia](#)" for additional details.

The signs and symptoms of the common cold are easily recognized. Since pharyngitis is often a concomitant symptom, it warrants focused clinical evaluation to exclude other etiologies. In the United States, pharyngitis accounts for approximately 6 to 10 million pediatric and >5 million adult healthcare visits annually.[15][16] While rhinovirus and adenovirus are the most prevalent etiologies of pharyngitis, many other viruses, bacteria, allergens, autoimmune, mechanical, and environmental irritants can cause overlapping symptoms.[17] The coexistence of rhinorrhea and cough is highly suggestive of a common cold etiology.

Examination of the pharynx should explore the presence and absence of erythema, exudates, foreign bodies, tonsillar hypertrophy, masses, and petechiae. The neck should be examined for the presence of adenopathy and thyroid masses. Group A beta-hemolytic *Streptococcus* is the most common bacterial etiology, occurring in 10% to 15% of adults and 15% to 30% of children with pharyngitis. Thus, testing for this pathogen should be considered when clinical features are suggestive.[15][16][18] Please see StatPearls' companion resources, "[Pharyngitis](#)" and "[Tonsillitis](#)," for additional details.

Evaluation of the ears should include an otoscopic examination to exclude the presence of otitis. Percussion and auscultation of the chest should be performed to evaluate the presence of reactive airways and lower respiratory tract involvement. Please see StatPearls' companion resources, "[Acute Otitis Media](#)" and "[Viral Pneumonia](#)," for additional details.

Evaluation

Approach to Upper Respiratory Infection Diagnostic Testing

Because a large variety of viral and bacterial pathogens are capable of causing overlapping symptoms, the diagnostic approach to URTIs can be a challenging and continuously moving target.[19][17] The majority of URTI cases are due to self-limited infection by respiratory viruses; regarding the common cold, the vast majority of cases will be due to rhinovirus and adenovirus, and do not require testing. However, seasonal factors, epidemics, pandemics, patient age, host immune factors, severity of illness, and examination findings can impact decisions on whether further diagnostic testing is indicated.

When considering which diagnostic modalities to obtain, clinicians must account for sensitivity, specificity, positive and negative predictive values, turnaround time, and resource utilization. Most importantly, deciding whether to obtain a test should rest on the probability that the result will significantly impact management and public health response. [20][21][22][23] Moreover, depending upon the setting (eg, home, outpatient clinic, emergency department, or inpatient ward) and the available resources, the type of diagnostic test available may be limited.

Over the past 20 years, a proliferation of rapid antigen and molecular tests have been developed for the diagnosis of URTIs. [24][17][22][20][23] Many of the tests are multiplex and capable of identifying various viral and bacterial pathogens, each having its own intra-test differences in sensitivity and specificity (see **Image. Pathogens Often Included in Multiplex Panels**). Many tests provide point-of-care results within minutes, while others require laboratory testing with variable turnaround times. Laboratories have changed the approach to testing algorithms, as has the advent of the COVID-19 pandemic. Seasonal epidemics of influenza and RSV can similarly impact the testing algorithm. [22][23] Adding to this diagnostic complexity is that asymptomatic colonization and prolonged viral shedding can result in positive tests that do not establish illness causality. Thus, a positive virus test panel may obfuscate the presence of a secondary bacterial coinfection. [20][17]

Numerous studies have concluded that tests for URTI are overutilized. Drivers of overutilization include ingrained practice habits, diagnostic uncertainty, reluctance to initiate a therapeutic regimen without a definitive diagnosis, clinician workload, and patient's desire for a diagnosis. A detailed discussion of the various diagnostic tests and best practice guidelines is beyond the scope of this course but may be obtained in several excellent reviews. [20][22][24][25][21]

All clinicians should consider the following questions before requesting a diagnostic test for adult and pediatric patients who present with clinical features of URTI to help guide decision-making:

- What pathogens are included in the diagnostic test panel?
- Should the test panel be modified to better reflect seasonal variations in circulating pathogens?
- What are the test's positive and negative predictive values?
- Does test sensitivity differ in children versus adults?
- How does the timing of the test relative to symptom onset impact its sensitivity?
- What is the test result turnaround time?
- Will the test result likely alter one's clinical impression?
- Will the test result change one's approach to management and therapy?
- Will the test result impact public health measures?
- How will the test impact workflow at the point of care and in the clinical microbiology laboratory?
- What does the test cost, and are there potential cost-saving offsets, such as decisions to discharge or admit patients, avoidance of ancillary testing, and avoidance of unnecessary antibiotics?
- What discomfort will the patient experience by obtaining the test specimen, and how should that be factored into the decision to obtain the test given the above questions?

Treatment / Management

The rhinovirus has no effective antiviral treatments. Therefore, therapy for the common cold is directed toward symptom relief. While hundreds of over-the-counter and behind-the-counter common cold "remedies" contain various ingredients, very few have undergone large randomized clinical trials, and even fewer have been extensively tested in children. Over-the-counter common cold medications should not be administered to children younger than 4 years due to safety concerns and lack of efficacy. In both adults and children, the use of common cold medications requires caution to prevent potential overdose, toxicity, and drug interactions. [26][27][CDC Common Cold Treatment]

Analgesics, eg, acetaminophen, nonsteroidal anti-inflammatory drugs, and topical anesthetics, can temporarily relieve sore throat pain. When formulated as a nasal spray, the anticholinergic medication ipratropium can reduce rhinorrhea symptoms. The antihistamines diphenhydramine and doxylamine can reduce rhinorrhea via their anticholinergic property; sedation is a common adverse effect. Adrenergic agents, eg, pseudoephedrine and oxymetazoline nasal spray, constrict the nasal venous sinuses, thereby reducing the nasal obstruction. Rebound congestion can develop when oxymetazoline nasal spray is used longer than 3 days.[28] Zinc lozenges may reduce cough and rhinitis symptoms when started within the first day of symptoms.[29]

Safety and efficacy concerns exist for the cough suppressants, dextromethorphan and codeine. When used alone or in combination with the expectorant guaifenesin, these cough suppressants have neither demonstrated efficacy in children nor definitive efficacy in adults. The safety and efficacy of the cough suppressant benzonatate have not been fully established; this medication is not recommended for children younger than 10.[26][27][30]

Cool mist humidifiers and sterile saline nasal drops and sprays can help moisten and clear nasal passages. Honey has antitussive properties but should not be used in children younger than 1 year of age due to the risk of botulism.[31] [32][CDC Common Cold Treatment][32] Mentholated rubs containing menthol, camphor, and eucalyptus may relieve cough and congestion. Irritation to the skin, eyes, and nose is a potential complication.[26] Additional details on the efficacy and safety of nonprescription, prescription, nonpharmacologic, and complementary therapies are beyond the scope of this course, but several excellent reviews on the subject are available.[26][27]

Antibiotics do not provide benefits in the treatment of the common cold. They are prescribed in approximately 30% to 40% of cases of self-limited URTIs and contribute significantly to the development of antibiotic resistance.[18][33] [34] In addition, unnecessary antibiotic use adds to medical costs and can produce significant adverse effects. There is often a desire by the patient to be placed on antibiotics for their URTIs. Antibiotic stewardship in this setting, while not always successful, requires effective communication skills and strategies to improve patient understanding of complex concepts. This is often a challenging but necessary exercise that requires the clinician to understand the risks associated with inappropriate antibiotic use. Antibiotics are indicated in the treatment of pharyngitis due to group A beta-hemolytic *Streptococcus* in severe cases of acute otitis media, in sinusitis lasting >10 days, or when sinus symptoms worsen after a period of initial improvement.[18]

Differential Diagnosis

URTIs have a broad spectrum of overlapping symptoms with variable onset and duration. The common cold is a syndrome comprising a constellation of symptoms with a relatively short onset. From a pragmatic standpoint, it is a clinical diagnosis. Common cold symptoms typically last from 7 to 10 days, although cough can linger for several weeks. The differential diagnoses is quite broad and a function of the sequence and severity of symptoms:

- Allergic rhinitis
- Bacterial pharyngitis
- Covid-19
- Influenza
- Epstein-Barr virus
- Acute HIV
- Acute bronchitis
- Pertussis

Toxicity and Adverse Effect Management

In children younger than 6 years, significant risks are associated with the administration of over-the-counter and prescription cough and cold medications. Most notably, overdose due to antihistamines and decongestants can be life-threatening. Similarly, the use of codeine should be avoided in the pediatric population. Inappropriate use of

antibiotics is a significant contributor to the emergence of antibiotic resistance, and they can cause significant adverse reactions.

Prognosis

The vast majority of URTIs are self-limited and mild to moderate in severity. Globally, in 2021, 12.8 billion episodes of URTIs and approximately 19,600 non-COVID-19 related deaths.[4] This represented a mortality rate of roughly 0.2 per 100,000 population. The highest mortality rates occurred in newborns and older adults. Geographically, the highest mortality rates occurred in sub-Saharan Africa.

Complications

Complications resulting from URTIs include the following:

- Lower respiratory infections
- Secondary bacterial infections
 - Otitis media
 - Mastoiditis
 - Sinusitis
 - Pneumonia
 - Exacerbation of asthma and COPD

Deterrence and Patient Education

Clinicians should understand the sensitivity, specificity, and positive and negative predictive values of the myriad rapid URTI tests available to them. Healthcare professionals should take the time to communicate with patients about the lack of efficacy and risks associated with administering antibiotics for the common cold.

Enhancing Healthcare Team Outcomes

Effective management of URTIs, particularly the common cold, requires a collaborative approach among physicians, advanced practitioners, nurses, pharmacists, and other healthcare professionals. Physicians and advanced practitioners must develop diagnostic skills to distinguish between viral and bacterial infections, ensuring that antibiotic stewardship principles are upheld. Nurses play a crucial role in patient education, reinforcing information about symptom management, medication risks, and the self-limiting nature of the common cold. Pharmacists contribute by counseling patients on the appropriate use of over-the-counter medications, identifying potential drug interactions, and advising on alternative non-pharmacologic treatments. By integrating their expertise, these professionals can prevent unnecessary prescriptions and reduce adverse drug effects, ultimately enhancing patient-centered care and safety.

Interprofessional communication is vital in aligning the care team's approach to managing URTIs. Physicians and advanced practitioners must clearly convey treatment plans to nurses and pharmacists, ensuring consistency in patient messaging. As frontline caregivers, nurses should relay patient concerns and adverse effects of medication to prescribers while pharmacists provide insights on drug safety and efficacy. Care coordination is further enhanced when healthcare professionals engage in shared decision-making, addressing patient expectations and misconceptions about treatment options. By fostering teamwork and maintaining open communication, healthcare professionals can improve patient adherence to evidence-based management strategies, reduce antibiotic misuse, and optimize both clinical outcomes and overall team performance.

Review Questions

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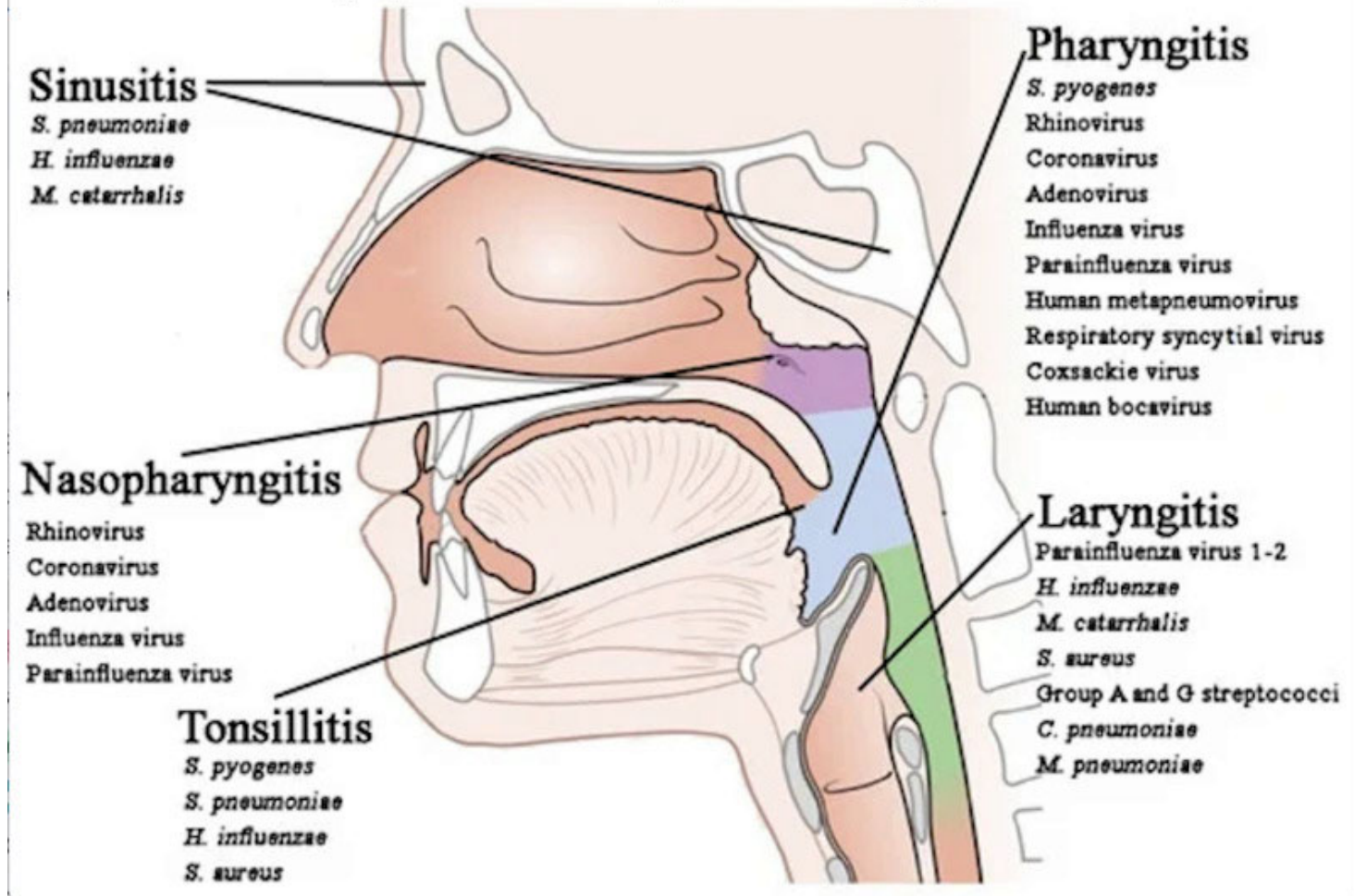
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Upper Respiratory Tract



Pathogens Often Included in Multiplex Panels. A proliferation of rapid antigen and molecular tests have been developed to diagnose URTIs, each with its own intra-test differences in sensitivity and specificity. Calderaro A, Buttrini M, Farina B, Montecchini S, DeConto F, Chezzi C. Respiratory tract infections and laboratory diagnostic methods: a review with a focus on syndromic panel-based assays. *Microorganisms*. 2022;10(9):1856. doi: [10.3390/microorganisms1009185684e](https://doi.org/10.3390/microorganisms1009185684e).

Pathogens included in usual respiratory viral panels

Influenza A

- H1 lineage
- H3 lineage
- pdm2009 H1 lineage

Influenza B

Adenovirus

HMPV

Rhinovirus or EV

RSV

Parainfluenza

- Types 1 to 4

Coronavirus

- OC43, NL63, 229E, HKU1

Bocavirus

Atypical bacterial pathogens occasionally included

Mycoplasma pneumoniae

Chlamydophila pneumoniae

Bordetella pertussis

Pathogens Often Included in Multiplex Panels. Contributed by R. Wunderink, MD

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